

TR Electronics Capstone

Vela Benedicto

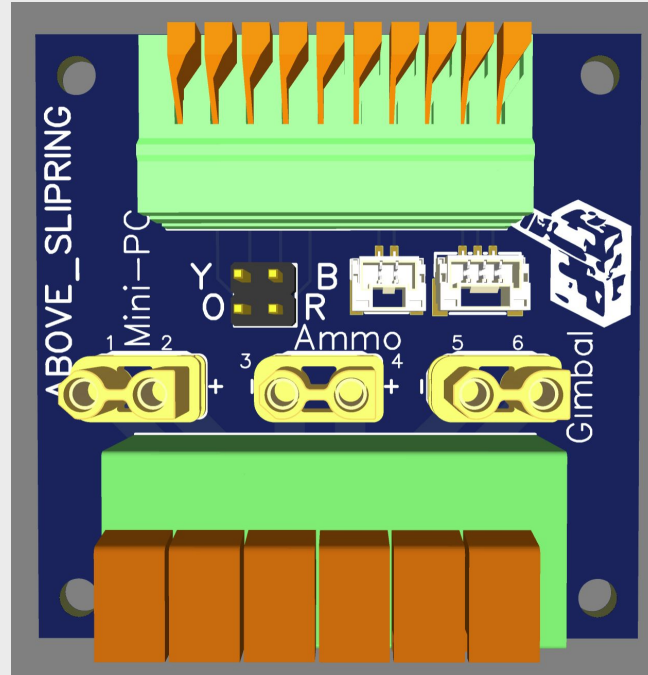
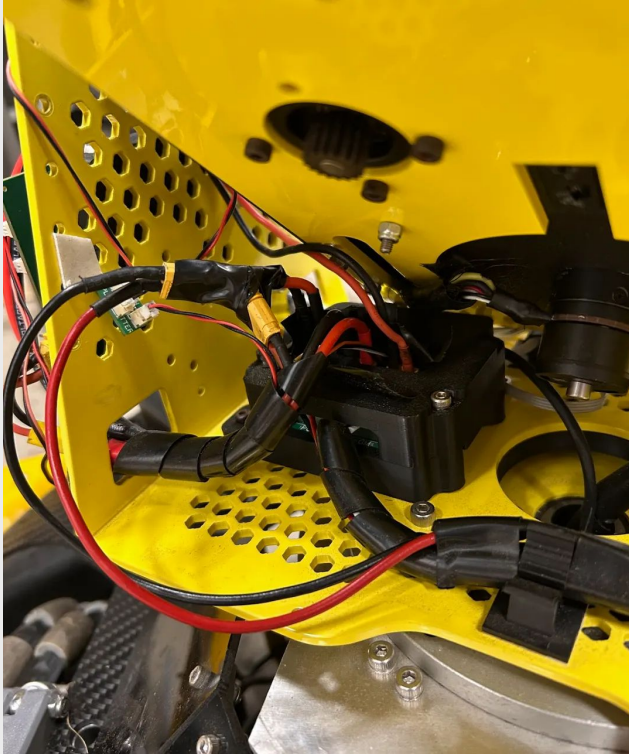


Slip ring



Slip ring: electromechanical device that transfers power, signals, and data between stationary and rotating parts

Current Method



Problem:

Insecure wires
(came out during
comp)



Objective and Requirements

Create a new slip ring breakout board that fulfills the following:

- **Power**

- **Mini-PC:** Y-split to 2 XT30s (nucleo and buck)
- **Gimbal:** to XT30
- **Ammo:** to XT30

Biggest constraint:
size & money

- **Signal**

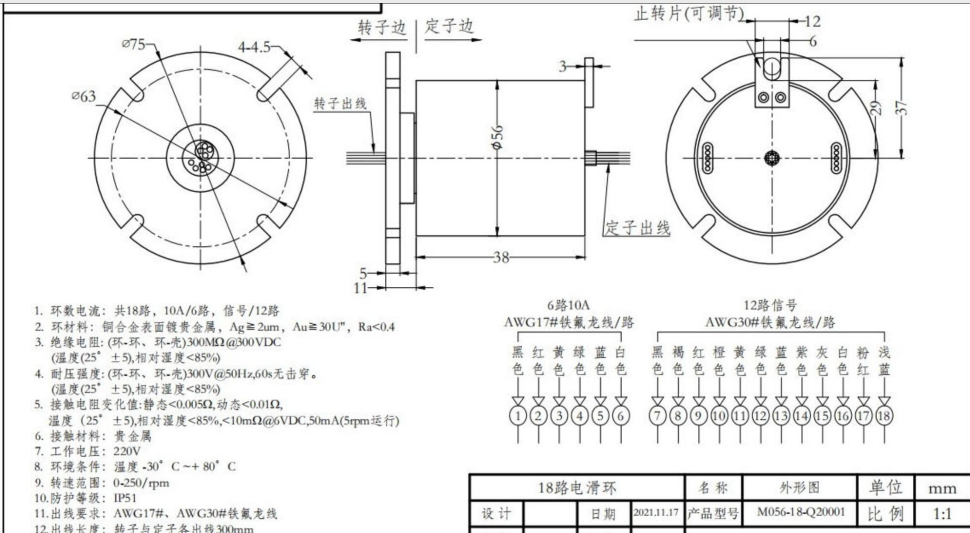
- Ensure wires are secure + signal integrity is maintained
- **CAN:** to JST GH 2p
- **Ref:** to JST GH 3p
- **VTM:** to aviator (M8 4p)
- Determine the best terminal connectors
- Consider the best wiring path for slip ring to reduce wire tension



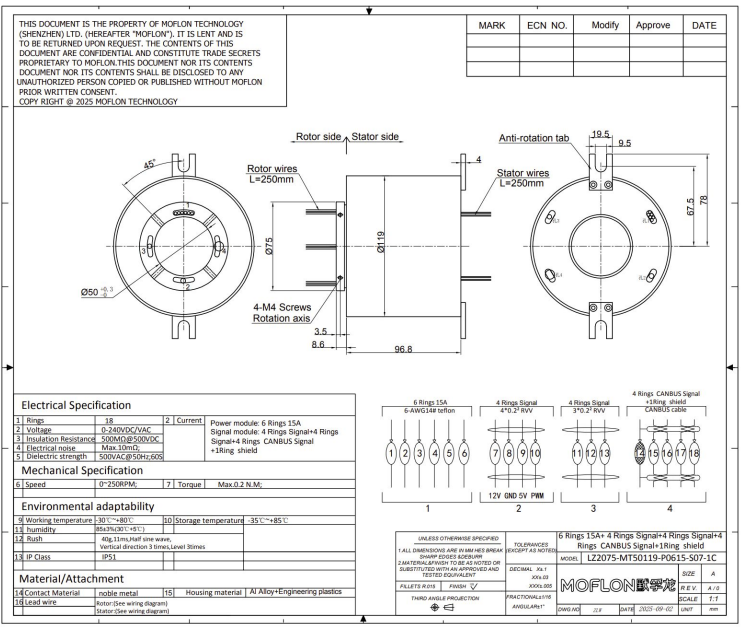
Datasheets

Standard series PART# Explanation

MT.Through Bore Slip Rings 50119-25-ID 25.4mm-119-OD 119mm	MT 50119 - P 06 10- S 06 - VC	Quality Level VC-Common(Popular)Level VD-Industrial Level Default=VC 06:6 Rings S:Signal(SA)
P:Power Rings 0610-06-6Ring-10-Rated 10A/Ring		



Senring



Moflon (hero)

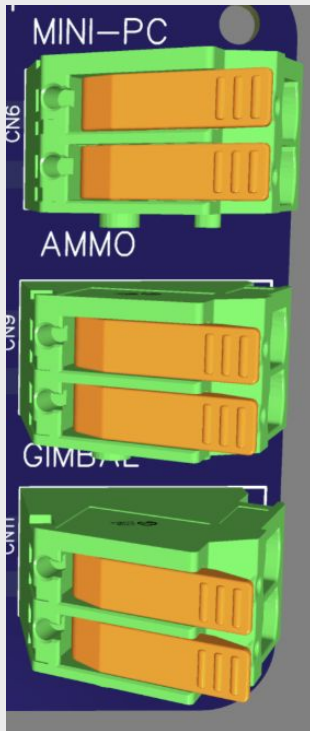


Datasheet Table

	Senring	Moflon (Hero)
POWER (6 wires)	17 AWG, 10A, Cu-tinned	14 AWG, 15A, Cu-tinned
SIGNAL	30 AWG, Cu-tinned	24 AWG, Cu
CAN (2 wires)		2 CAN
Nucleo (3 wires)		RX, TX, GND
Aviator (4 wires)		2 CAN, 12V, GND
	+3 wires extra	+2 wires extra (5V, PWM)

Power Connectors (6 wires)

- Phoenix contact connectors
 - Ideal: 6p or 2p, 10A, 12V, lever, $P = 5.08\text{mm}$
 - Famous for high-quality terminal blocks
 - Efficient, versatile, durable

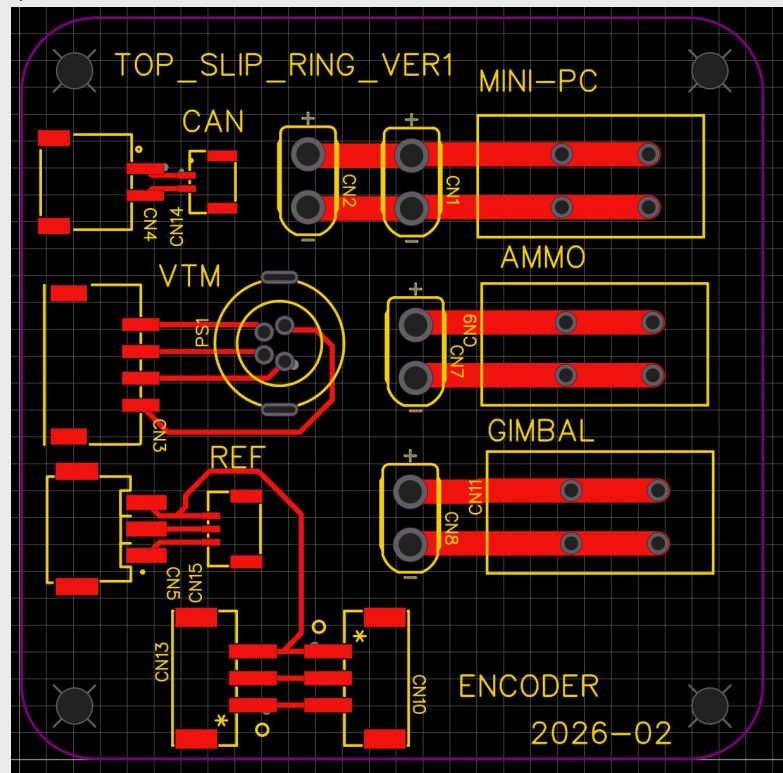
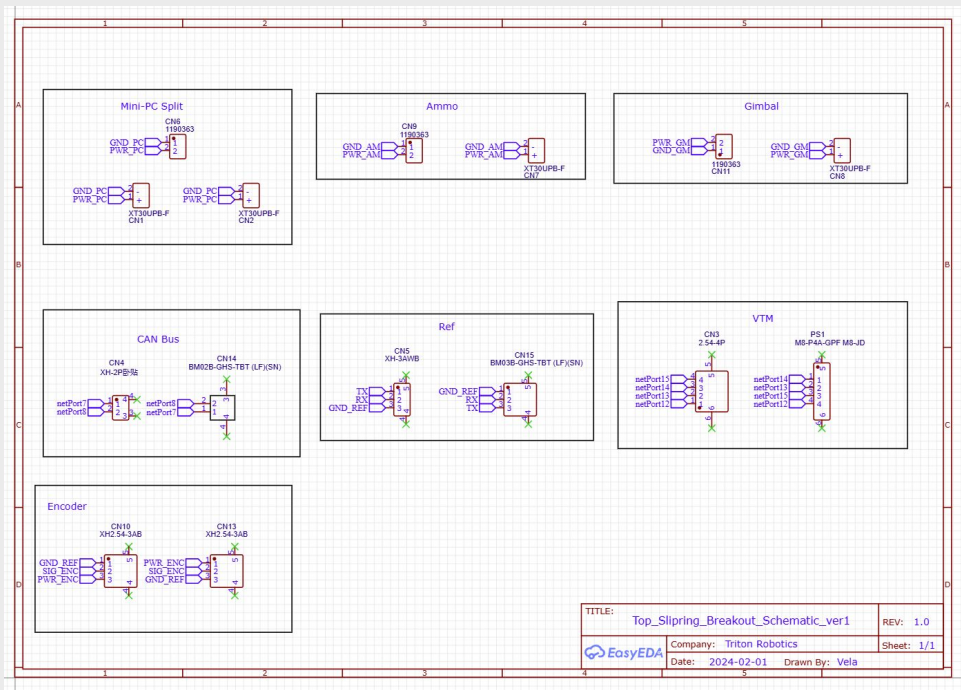




JST Connectors (11-12 signal wires)

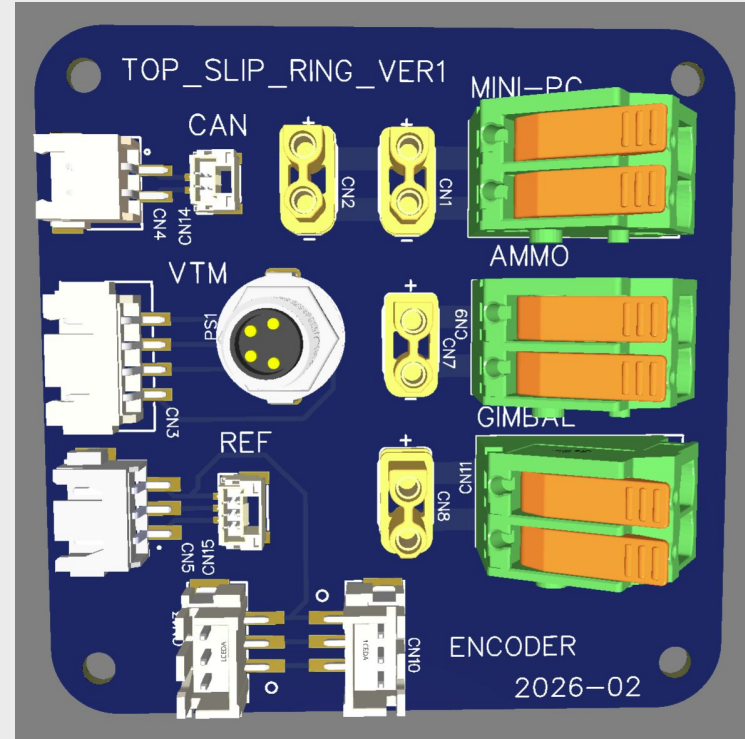
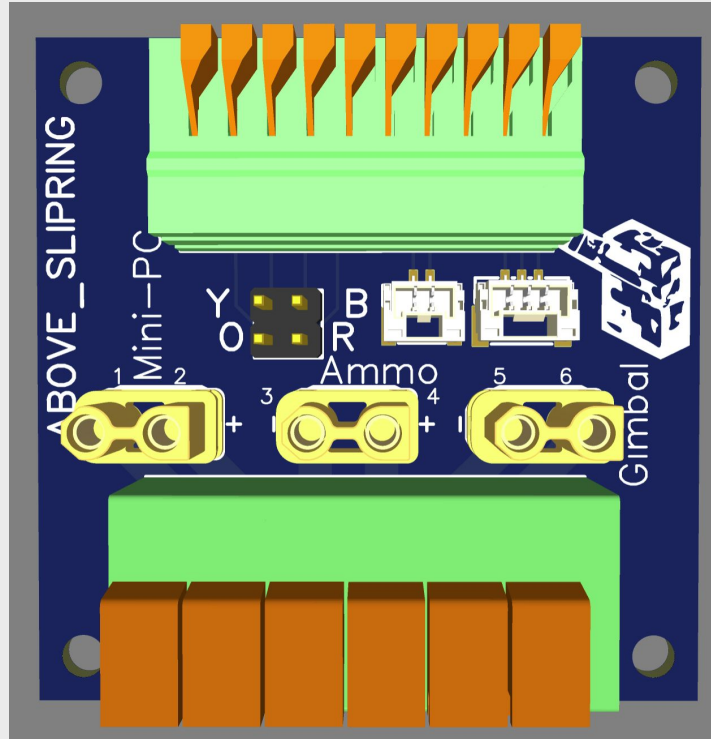
Connector	Pitch (mm)	AWG	Locking
XH	2.54	20-28	Y (friction)
GH	1.25		Y (latching)
PA	2.0		Y (friction fit)
PH	2.0		Y (friction fit)
VH	3.96		Y (latching)
SM	2.5		Y (latching)

Schematic & PCB (70x70mm)





Final :))



Buck Y-split for LiDAR + Jetson

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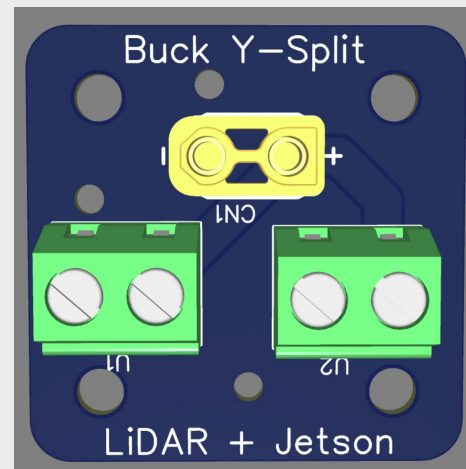
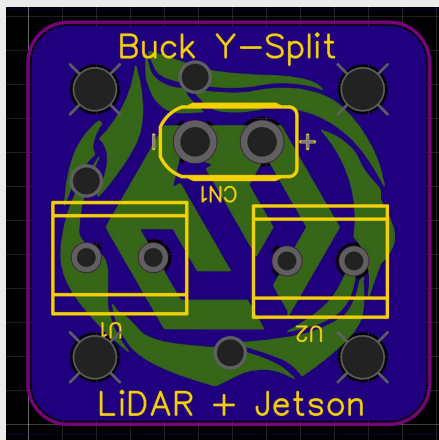
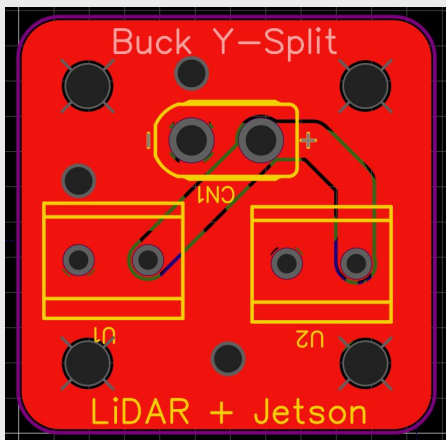
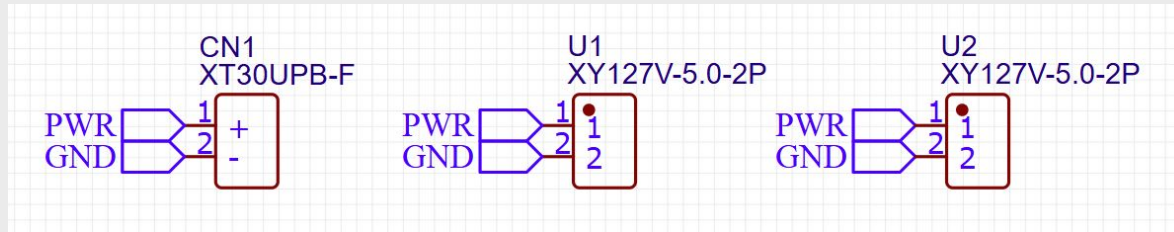




Objective and Requirements

- Create a board that goes from the buck converter (XT30) to 2 ends:
 - to Jetson (24AWG)
 - to LiDAR (unitree I2 = 24AWG, odin 1 = 24AWG)
- Jetson draws 45W, 12V and LiDAR draws 10W, 12V

Schematic, PCB, Final (30x30mm)





Questions? :))